E40: the chain solve's Macaulay matrix stored from each row's lead

Run and results report, 2026-11-02. Research note `research/notes/index-calculus/RESEARCH_GLV_INVARIANT_FACTOR_BASES.md`: §9.50 (pre-registration, commit `5a0dd20af9`, after the Stage 0 count and before any solver code) and §9.51 (results). Code: `64a385da03` (Stage 0 instrument), `0787278d7b` (`src/cryptanalysis/macaulay_profile.rs`), `f2f31dbec1` (`klein_v4` hook, committed alone), `d6d00182c5` (record fields), `9b2a47238b` (harness), `86d0f86c25` (controls fix). Binaries built from clean, blob-checked exports and pinned by sha256 (`experiments/47_macaulay_entries/binary_sha256.txt`, `binary_commit.txt`). Records: `experiments/47_macaulay_entries/`. Host: the 4-vCPU container, shared with two other experiments (load about 17); isolab-eks unreachable, `conductor` not installed. **Operation counts only, no wall-time claim.** Python only plots recorded values; every number below comes from `examples/e40_stage0.rs`, `e40_stage_a.rs`, `e40_checks.rs` and `e40_tables.rs`.

Outcome

- **Stage 0 (counts).** Of the 176,064 cells per call of the degree-11 Macaulay matrix (393×448), 73,237 (41.6 %) lie left of each row's lead. A forward echelon by lead and the left-looking normal forms never read them; only the dense first lead scan does (73,630 of 79,115 lead-scan reads). There are no duplicate rows and no untouched columns. 9 rows reduce to zero, holding 434 cells. One border target of degree 11 occurs per call, so $d = 11$ is forced.
- **Lever V (`chain-profile`).** Each row is stored from its lead. The arithmetic, pivots and outputs are A's. Every use of an offset is charged as a native operation (about 22,400 per call).
- **Identity holds everywhere.**
  - 7/7 existing configurations are byte-identical.
  - Stage A: 10,000/10,000 calls (outputs, fallback flag, every phase count, pivot digest, stored-cell counter, lead-scan relation).
  - End to end: 42/42 records equal A's (or A''s) outside the tag, the timing fields, the lead scans and the two E40 fields.
- **Controls.**
  - P1 (lead cell dropped) was detected on 320/320 calls.
  - P2 (counter short by one per row) was detected on 320/320 calls.
  - The perturbed coefficient changed 213/213 non-empty solution sets.
  - The negative control A′ (A on another stream) is not ranked against A. V′ against A′ replicates the bands.
- **Band, 32 rows.**
  - Against A: S_A/S_V is 1.0000 / 1.2518–1.2618 / 1.3629–1.3807 at 0 / 0.1 / 1 GAE.
  - Against E25: S_{E25}/S_V is 2.5054–2.5766 / 1.7455–1.8033 / 1.4105–1.4558.
  - **Ranked by §9.50′s rule:** V is equal to A at 0 GAE (no improvement there), ahead of A at 0.1 and 1 GAE, and ahead of E25 at every weight.
  - Every one of G2.204–G2.215 holds.
- **Distance to rho** (walk (iii)): rho is ahead of every row at every weight. The best row is GLS $p = 397$ seed 1 arm B, with S_V/ρ = $6.00 \cdot 10^3$ / $1.58 \cdot 10^4$ / $6.17 \cdot 10^4$ (A: $6.00 \cdot 10^3$ / $1.99 \cdot 10^4$ / $8.51 \cdot 10^4$).

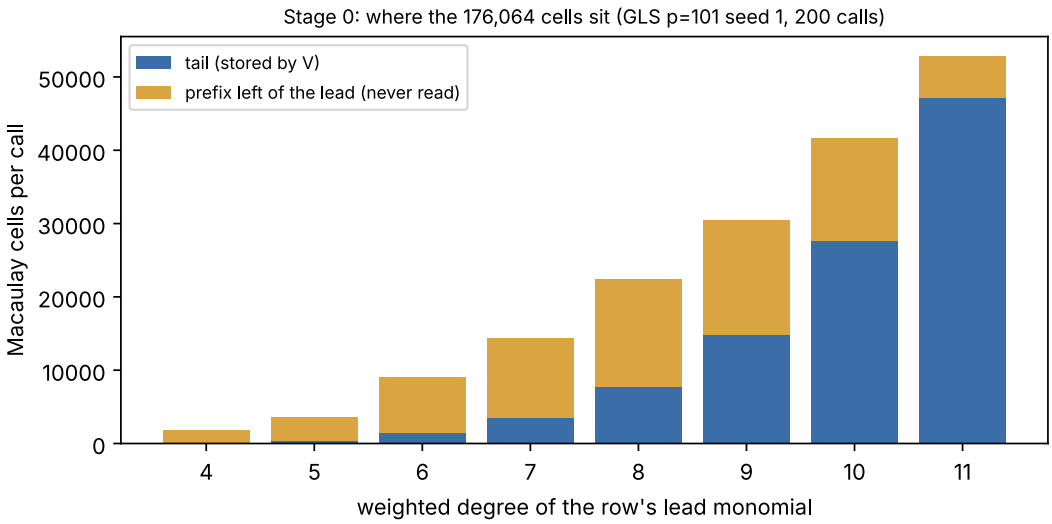


Figure 1: Stage 0 (`experiments/47_macaulay_entries/stage0.json`, GLS $p = 101$ seed 1, 200 calls): Macaulay cells per call by the weighted degree of the row's lead. The prefix (gold) is never read by the echelon or the normal forms; V stores only the tail (blue). The other 21 curves give the same split to within one cell per call.

Stage 0: where the native work of the build goes (per call, 22 curves × 200 calls)

part	count
columns / rows built / shift candidates / skipped by row selection	448 / 393 / 480 / 87
monomials enumerated (and sorted) / selection tests / index inserts	928 / 659 / 448
coefficient writes = column hash lookups	11,154–11,251
dense cells `rows · cols` (priced by the band)	176,064
prefix left of the lead / tail	73,237–73,239 / 102,825–102,827
first lead scan / all lead-scan reads (A)	73,630–73,632 / 79,114–79,116
zero rows (cells) / duplicate rows / untouched columns	9 (434) / 0 / 0
rows in the normal forms' dependency closure (others' tails)	334 (16,105–16,124)

The other candidate levers are not proposed: zero rows (0.25 % of cells, unknown before the echelon; E21 closed the trace route), rows outside the dependency closure (they fix the staircase; a trace-shaped lever), degree truncation ($d = 11$ forced), per-curve reuse (moves no priced work), and V_4/ψ redundancy (the system is already invariant). Lead atlas: no lead covers this build. F4 trace reuse is closed (binary); F5 signatures and prime-IC constant engineering are open.

Stage A (10 curves × 1,000 fresh calls, per-call means)

quantity	result (prediction)
V stored cells, clean calls	102,826–102,827 ($[1.02, 1.04] \cdot 10^5$)
cells A / V, clean calls	1.7122–1.7123 ([1.69, 1.73])
V offset operations	22,340–22,489 ($[1.5, 3.0] \cdot 10^4$)
lead-scan reads A / V	79,114–79,117 / 5,877–5,879
S_A/S_V at 0 / 0.1 / 1 GAE	1.000000 / 1.2447–1.2658 / 1.3534–1.3831
σ1 (lead scans priced) at 0.1 / 1 GAE	1.5881–1.6305 / 1.8389–1.8968

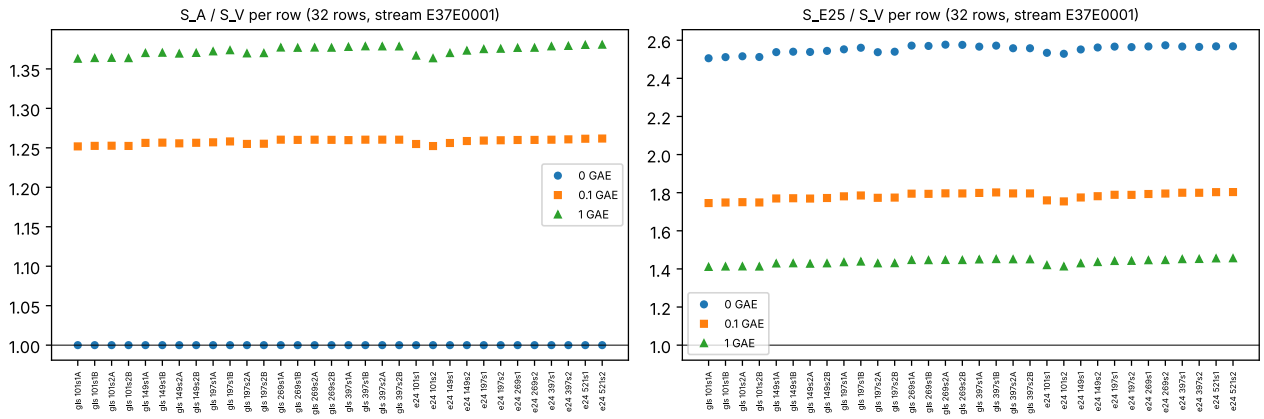


Figure 2: End to end, 32 rows (20 GLS, 12 E24) on stream `0xE37E0001`, paired with E37′s A and E25 records: S_A/S_V (left) and S_{E25}/S_V (right) at 0, 0.1 and 1 GAE (1 GAE = 97 F_p multiplications). Source: `experiments/47_macaulay_entries/summary.json`.

End to end (32 rows, Karatsuba, paired)

quantity	0 GAE	0.1 GAE	1 GAE
S_A/S_V	1.0000	1.2518–1.2618	1.3629–1.3807
S_{E25}/S_V	2.5054–2.5766	1.7455–1.8033	1.4105–1.4558
σ1 S_A/S_V	1.0000	1.6024–1.6217	1.8572–1.8921
σ4 S_A/S_V	1.0000	1.2362–1.2449	1.3312–1.3461
$S_V/$ rho	$6.00 \cdot 10^3$–$1.98 \cdot 10^5$	$1.58 \cdot 10^4$–$5.41 \cdot 10^5$	$6.17 \cdot 10^4$–$2.36 \cdot 10^6$
$S_A/$ rho	$6.00 \cdot 10^3$–$1.98 \cdot 10^5$	$1.99 \cdot 10^4$–$6.80 \cdot 10^5$	$8.51 \cdot 10^4$–$3.24 \cdot 10^6$

Negative control on stream `E40E0002` (20 GLS rows): A′ against A is ahead on 10 rows and behind on 10 (not ranked). V′ against A′ is 1.0000 / 1.2506–1.2608 / 1.3613–1.3790. The per-row table is in §9.51 and `tables.md`.

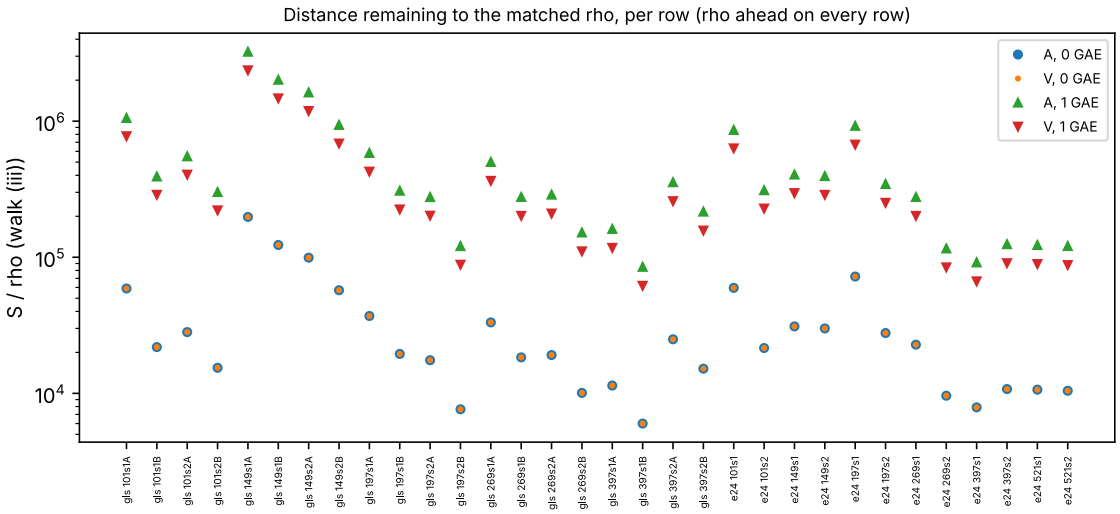


Figure 3: Distance remaining to the matched rho (walk (iii)), per row, for A and V at 0 and 1 GAE (at 0 GAE they coincide). Rho is ahead on every row. Source: `summary.json`.

Deviations and limits

- The first controls harness compared the decomposition sets after the lift. Those were empty on all 320 calls, so the perturbed-coefficient control was vacuous. §9.50 cites E31′s control, which is defined on solution sets. The harness was corrected (`86d0f86c25`, binary `e40_checks_v2`) and rerun before any verdict. Both runs are kept (`checks_v1.*`, `checks.*`).
- A falling-back call is charged A's convention (2 · rows · cols), so V gains nothing on those calls (conservative).

- At 0 GAE, V and A are identical. The improvement exists only at the band's positive weights.
- Scope: E25′s and E24′s curves, $p \leq 521$, degree 11, one host. The 176,064-cell figure for A is the registered model. No wall-time claim is made.
- The validation gate, the leaderboard and `benchmarks/` are left to the main session.